

JEE MAIN REVISION MASTER: CHEMISTRY IN EVERYDAY LIFE OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 77 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Chemistry In Everyday Life - Core Application	77 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Chemistry In Everyday Life - Core Application (Very High Yield — 77 PYQs)

Core Principle: Key problem pattern in Chemistry In Everyday Life.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- **[2024 Morning Shift] Q#1 (MCQ):** "Match List I with List II LIST I (Compound) LIST II (Uses) A. Iodoform I. Fire extinguisher B. Carbon tetrachloride II. Insecticide C. CFC II..." **[Ans: B]**
- **[2024 Evening Shift] Q#2 (MCQ):** "Match List - I with List - II. List I (Compound) List II (Use) (A) Carbon tetrachloride (I) Paint remover (B) Methylene chloride (II) Refrig..." **[Ans: N/A]**
- **[2023 Morning Shift] Q#1 (MCQ):** "Match List I with List II: LIST I LIST II A. Saccharin I. High potency sweetener B. Aspartame II. First artificial sweetening agent C. Alita..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2024 (Online) 6th April Morning Shift (Q#1)

Match List I with List II LIST I (Compound) LIST II (Uses) A. Iodoform I. Fire extinguisher B. Carbon tetrachloride II. Insecticide C. CFC III. Antiseptic D. DDT IV. Refrigerants Choose the correct answer from the options given below:

- (A) -II, B-IV, C-I, D-III
- (B) -III, B-I, C-IV, D-II **[Correct]**
- (C) -III, B-II, C-IV, D-I
- (D) -I, B-II, C-III, D-IV

Correct Answer: B

Solution Analysis: The matching for each compound with its respective use is based on their known applications: Iodoform is a compound known for its application as an antiseptic . It has been used historically in medical applications for the treatment of wounds due to its antimicrobial properties. Hence, Iodoform matches with III. Antiseptic . Carbon Tetrachloride was widely used as a fire extinguishing agent due to its property of being a non-flammable liquid that can suffocate fires. However, its use has been significantly reduced in most countries because of its adverse health and environmental impacts. Carbo